

Aim: To implement DML commands using clauses, operators and functions in series.

Data Manipulation language (DML)

The DML is used to retrieve, insert and modify database information. These commands will be used by all database users during the routine operation of the database.

DML Commands:

1. Insert INTO: This is used to add records into a relation.

Syntax: insert into <tablename> (field1, field2, ..., fieldn)
values (data-1, data-2, ..., data-n);

Example:

SQL> insert into customer values (238, 'Ram', 'chennai', '892468');

SQL> insert into customer values (409, 'Rocky', 'Vizag', '8449088');

SQL> insert into customer values (112, 'visat', 'Hyderabad', '7049867');

After inserting:

Customer ID	name	address	Ph-no
238	Ram	Chennai	892468
409	Rocky	Vizag	8449088
112	visat	Hyderabad	7049867

Update - Set-where

This is used to update the content of a record in a relation.

Syntax:

SQL> Update relation name set field-name1 = data, field-name2 = data, where field-name = data;

Example:

SQL> update customer set name = "kumar" where customer ID = 409

After updating

Customer ID	name	address	Ph-no
238	Ram	Chennai	892468
409	Kumar	Vizag	8449085
112	Virat	Hyderabad	7049867

3. Delete - from:

a) This is used to delete all the records of a relation but it will retain the structure of that relation.

Syntax: SQL > Delete from table-name;

Ex:- SQL Delete from customers;

After deleting:

Customer-ID	name	address	Ph-no.

b) Delete - from.where: This is used to delete a selected record from a relation.

Syntax: SQL > Delete from relation-name where condition;

Ex:-

SQL > Delete from customers where name = 'Ram';

After deleting:

Customer-ID	name	address	Ph-no.
409	Kumar	Vizag	8449085
112	Virat	Hyderabad	7049867.

5. Truncate

This command will remove the data permanently. But structure will not be removed.

Syntax: Truncate Table <table name>

Ex: Truncate Table Customers;

After truncate:

Customer ID	name	address	Ph-no

Queries

1. Retrieve a member name starts with letter 'V'.

Query: Select name from bank-account where name like 'V.%';

Output:

Name

Vijay

Vikram

Virat

2. List of Accounts where balance between 10000 and 20000;

Query: Select * from bank-account where balance between 10000 and 20000;

Output:

Name	Account-number	Balance	Category
Vijay	2345	10000	Savings
Vikram	7890	20000	Savings

3. Finding records who has minimum Balance

Query: Select min(balance) from bank-account;

Output:- min(Balance)
10000

4. Finding records who has Balance ≥ 20000 ;

Query:- Select * from bank-account where balance ≥ 20000

Output:

Name	Account-number	Balance	Category
Vikram	7890	20000	Savings
Virat	4567	35000	Salary
Akash	8987	50000	RD

5. Distinct

Query:- Select distinct category from Bank-account;

Output: Category
Savings
Salary
RD

6. Union :

Query:- Select name from Customer union Select name from bank account;

Output: Name
Rocky
Vishat
Vijay
Vikram
Akash

VEL TECH	
EX NO.	3.1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	15
TOTAL (20)	40
SIGN WITH DATE	24/8

Result: The implementation of DML commands using clauses, operators and functions in queries executed successfully.

Date: 26/8/25

Aggregate Functions

Aim:- To study and implement aggregate functions
(count(), sum(), avg(), min(), max()).

Procedure:

1. Create a table named Bank-Account.
2. Insert sample records
3. write queries using aggregate functions.
4. Observe and record the output.

commands with explanation.

1. Count the total number of students,

Select Count * AS total-amount from bank-Account;

Output:- Total-amount
4

2. Find the highest amount in the account.

Select max(balance) AS highest-amount from
bank-account;

Output :- highest-amount
50000

3. Find the average amount of accounts.

Select Avg(balance) AS average-amount from
bank-account;

Output:- Average-amount
28750.

4. Find minimum Amount of the Account

Query:- Select min (balance) as min-amount from
Bank-account;

Output:- min-account
10000

5) Find the total amount in the Bank Account in each category.

Query:- Select category, sum(balance) as total.amount
from bank-account group by category;

Output:-

Category	Total-Amount
Savings	30000
Salary	35000
RD	50000

6. Find the average Balance per category ordered by average balance descending.

Query:- Select category, avg(balance) as avg-balance from
bank-account group by category order by avg-
balance desc;

Output: Category- Avg-Balance

RD	10000
Salary	35000
Savings	15000

VEL TECH	
EX NO.	2.2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (3)	5
VIVA VOCE (5)	5
RECORD (5)	15
TOTAL (20)	40
SIGN WITH DATE	26/8

Result: The implementation of aggregate functions
executed successfully.